Copyrighted Materials
Copyright © 1994 Oxford University Press Retrieved from www.knovel.com

2

BINARY ELASTIC COLLISIONS

2.1 Momentum and energy considerations

We have seen that intermolecular collisions in dilute gases are overwhelmingly likely to be binary collisions involving just two molecules. An elastic collision is defined as one in which there is no interchange of translational and internal energy. The pre-collision velocities of the two collision partners in a typical binary collision may be denoted by $\boldsymbol{c}_1$ and $\boldsymbol{c}_2$. Given the physical properties of the molecules and the orientation of the trajectories, our task is to determine the post-collision velocities $\boldsymbol{c}_1^*$ and $\boldsymbol{c}_2^*$.

Linear momentum and energy must be conserved in the collision. This requires

$$m_1\boldsymbol{c}_1 + m_2\boldsymbol{c}_2 = m_1\boldsymbol{c}_1^* + m_2\boldsymbol{c}_2^* = (m_1+m_2)\boldsymbol{c}_\mathrm{m} \tag{2.1}$$

and

$$m_1 c_1^2 + m_2 c_2^2 = m_1 c_1^{*2} + m_2 c_2^{*2}. \tag{2.2}$$

Here, m_1 and m_2 are the masses of the two molecules and $\boldsymbol{c}_\mathrm{m}$ is the velocity of the centre of mass of the pair of molecules. Eqn (2.1) shows that this centre of mass velocity is not affected by the collision. The pre-collision and post-collision values of the relative velocity between the molecules may be defined by

$$\boldsymbol{c}_\mathrm{r} = \boldsymbol{c}_1 - \boldsymbol{c}_2$$

and (2.3)

$$\boldsymbol{c}_\mathrm{r}^* = \boldsymbol{c}_1^* - \boldsymbol{c}_2^*.$$

Eqns (2.1) and (2.3) may be combined to give

$$\boldsymbol{c}_1 = \boldsymbol{c}_\mathrm{m} + \frac{m_2}{m_1+m_2}\boldsymbol{c}_\mathrm{r}$$

and (2.4)

$$\boldsymbol{c}_2 = \boldsymbol{c}_\mathrm{m} - \frac{m_1}{m_1+m_2}\boldsymbol{c}_\mathrm{r}.$$

The pre-collision velocities relative to the centre of mass are $\boldsymbol{c}_1 - \boldsymbol{c}_\mathrm{m}$ and $\boldsymbol{c}_2 - \boldsymbol{c}_m$. Eqn (2.4) shows that these velocities are antiparallel in this frame of reference and, if the molecules are point centres of force, the force

between them remains in the plane containing the two velocities. The collision is therefore planar in the centre of mass frame. The post-collision velocities may similarly be obtained from eqns (2.1) and (2.3) as

$$\boldsymbol{c}_1^* = \boldsymbol{c}_m + \frac{m_2}{m_1+m_2}\boldsymbol{c}_r^*$$

and (2.5)

$$\boldsymbol{c}_2^* = \boldsymbol{c}_m - \frac{m_1}{m_1+m_2}\boldsymbol{c}_r^* .$$

This shows that the post-collision velocities are also antiparallel in the centre of mass frame. The conservation of angular momentum requires that the projected distance between the post-collision velocities be equal to the projected distance b between the pre-collision velocities.

Eqns (2.4) and (2.5) show that

$$m_1 c_1^2 + m_2 c_2^2 = (m_1+m_2)c_m^2 + m_r c_r^2$$

and (2.6)

$$m_1 c_1^{*2} + m_2 c_2^{*2} = (m_1+m_2)c_m^2 + m_r c_r^{*2},$$

where

$$m_r = \frac{m_1 m_2}{m_1+m_2} \tag{2.7}$$

is called the *reduced mass*. A comparison of eqn (2.6) with the energy equation (2.2) shows that the magnitude of the relative velocity is unchanged by the collision, i.e.

$$c_r^* = c_r . \tag{2.8}$$

Since both $\boldsymbol{c}_m$ and $\boldsymbol{c}_r$ may be calculated from the pre-collision velocities, the determination of the post-collision velocities reduces to the calculation of the change in direction χ of the relative velocity vector.

If $\boldsymbol{F}$ is the force between two spherically symmetric point centre of force molecules and $\boldsymbol{r}_1, \boldsymbol{r}_2$ are their position vectors, the equations of motion of the molecules are

$$m_1 \ddot{\boldsymbol{r}}_1 = \boldsymbol{F}$$

and (2.9)

$$m_2 \ddot{\boldsymbol{r}}_2 = -\boldsymbol{F}.$$

Hence

$$m_1 m_2 (\ddot{\boldsymbol{r}}_1 - \ddot{\boldsymbol{r}}_2) = (m_1+m_2)\boldsymbol{F}$$

or, if the relative velocity vector is denoted by $\dot{\boldsymbol{r}}$,

$$m_r \ddot{\boldsymbol{r}} = \boldsymbol{F}. \tag{2.10}$$

The motion of the molecule of mass m_1 relative to the molecule of mass m_2 is therefore equivalent to the motion of a molecule of mass m_r relative to a fixed centre of force.

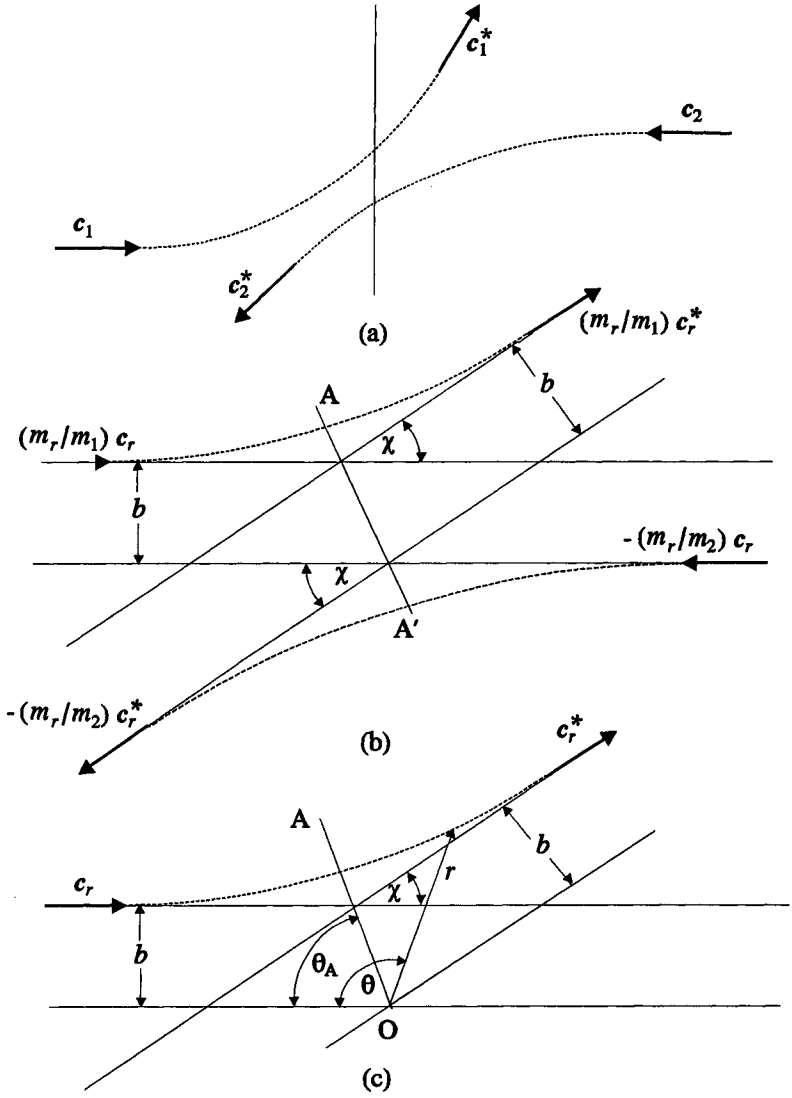


Fig. 2.1 Frames of reference for the analysis of binary collisions.
(a) Planar representation of a collision in the laboratory frame of reference.
(b) Binary collision in the centre of mass frame of reference.
(c) Interaction of the reduced mass particle with a fixed scattering centre.

The above results are summarized in Fig. 2.1. The transformation from the laboratory to the centre of mass coordinate system transforms the three-dimensional trajectories into two-dimensional trajectories that are symmetrical about the *apse line* AA′. The two trajectories are reduced to one in the further transformation to the reduced mass frame of reference, and this trajectory remains symmetrical about the transformed apse line which passes through the scattering centre O. This symmetry reflects the symmetry of the equations with respect to the pre-collision and post-collision velocities. A further consequence of this symmetry becomes apparent if we consider a collision between two molecules of velocity $\boldsymbol{c}_1^*$ and $\boldsymbol{c}_2^*$ and such that the separation of their undisturbed trajectories in the centre of mass frame of reference is again equal to b. This collision results in post-collision velocities of $\boldsymbol{c}_1$ and $\boldsymbol{c}_2$ and is called the *inverse* of the original or *direct* collision. The trajectories of the direct and inverse collisions in the reduced mass frame of reference are shown in Fig. 2.2.

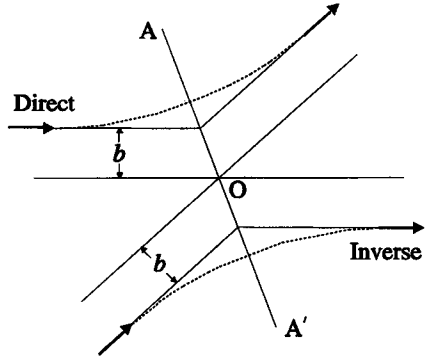


Fig. 2.2 Representation of the direct and inverse encounters of a reduced mass particle with a fixed scattering centre.

2.2 Impact parameters and collision cross-sections

Apart from the translational velocities of the two collision partners, just two *impact parameters* are required to completely specify a binary elastic collision between spherically symmetric molecules. The first is the distance of closest approach b of the undisturbed trajectories in the centre of mass frame of reference. The plane in which the trajectories lie in the centre of mass frame is called the collision plane, and the second impact parameter is chosen as the angle ε between the collision plane and some reference plane. As shown in Fig. 2.3, the line of intersection of the collision and reference plane is parallel to $\boldsymbol{c}_r$. If we now consider the plane normal to $\boldsymbol{c}_r$ and containing O, the *differential cross-section* $\sigma d\Omega$ for the collision specified by the impact parameters b and ε is defined by

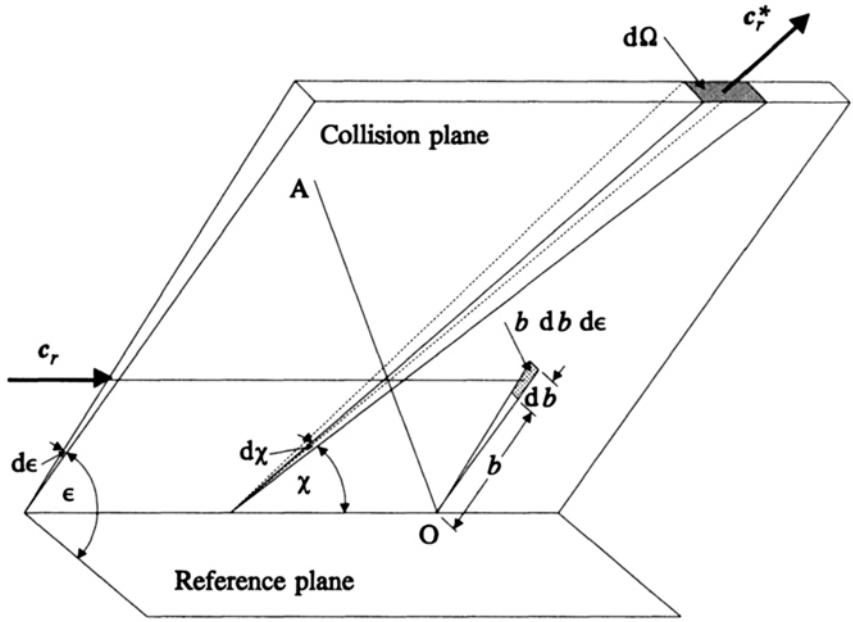


Fig. 2.3 Illustration of the impact parameters.

$$\sigma\, d\Omega = b\, db\, d\varepsilon\,, \tag{2.11}$$

where $d\Omega$ is the unit solid angle about the vector $\boldsymbol{c}_r^*$. From Fig. 2.3

$$d\Omega = \sin\chi\, d\chi\, d\varepsilon \tag{2.12}$$

so that

$$\sigma = (b/\sin\chi)\,|\,db/d\chi\,|\,. \tag{2.13}$$

Finally, the *total collision cross-section* σ_T is defined by

$$\sigma_T = \int_0^{4\pi} \sigma\, d\Omega = 2\pi \int_0^{\pi} \sigma \sin\chi\, d\chi\,. \tag{2.14}$$

We will find that this integral diverges for some of the more realistic molecular models and it is then necessary to introduce effective or nominal cross-sections.

2.3 Collision dynamics

With reference to the polar coordinates r, θ defined in Fig. 2.1(c), the angular momentum and energy of the particle of reduced mass m_r within the force field about O may be equated to the limiting values of these quantities as $r \to \infty$. The equation for the angular momentum is simply

$$r^2\dot{\theta} = \text{const} = bc_r. \tag{2.15}$$

The energy within the force field is the sum of the kinetic and potential energies and, since the force tends to zero at large distances, this may be equated to the asymptotic kinetic energy, i.e.

$$\tfrac{1}{2}m_r(\dot{r}^2 + r^2\dot{\theta}^2) + \phi = \text{const} = \tfrac{1}{2}m_r c_r^2 = E_t. \tag{2.16}$$

Here, ϕ is the intermolecular potential which is related to the spherically symmetric force $\boldsymbol{F}$ between the molecules by

$$\phi = \int_r^\infty F\,dr, \tag{2.17}$$

or

$$F = -d\phi/dr.$$

Time may be eliminated from eqns (2.15) and (2.16) to give the equation of the orbit as

$$\left(\frac{dr}{d\theta}\right)^2 = \frac{r^4}{b^2} - r^2 - \frac{\phi r^4}{\tfrac{1}{2}m_r c_r^2 b^2}.$$

Introducing the dimensionless coordinate

$$W = b/r, \tag{2.18}$$

this becomes

$$(dW/d\theta)^2 = 1 - W^2 - \phi/(\tfrac{1}{2}m_r c_r^2),$$

so that

$$\theta = \int_0^W \{1 - W^2 - \phi/(\tfrac{1}{2}m_r c_r^2)\}^{-\frac{1}{2}}\,dW.$$

Now at the intersection of the orbit with the apse line OA,

$$\theta = \theta_A$$

and

$$dW/d\theta = 0.$$

Therefore

$$\theta_A = \int_0^{W_1} \{1 - W^2 - \phi/(\tfrac{1}{2}m_r c_r^2)\}^{-\frac{1}{2}}\,dW, \tag{2.19}$$

where W_1 is the positive root of the equation

$$1 - W^2 - \phi/(\tfrac{1}{2}m_r c_r^2) = 0. \tag{2.20}$$

Finally, the deflection angle of the relative velocity is

$$\chi = \pi - 2\theta_A. \tag{2.21}$$

The above determination of an expression for the deflection angle χ constitutes the key step in the analysis of the dynamics of a binary elastic collision. However, as noted at the beginning of the chapter, the objective is to determine the post-collision velocities $\boldsymbol{c}_1^*$ and $\boldsymbol{c}_2^*$. In a typical calculation, the components and magnitudes of $\boldsymbol{c}_m$ and $\boldsymbol{c}_r$ would be obtained from the pre-collision velocities of the collision partners through eqns (2.1) and (2.3). The specification of the impact parameter b then allows the deflection angle χ to be calculated from eqns (2.18) – (2.21). The components of $\boldsymbol{c}_r^*$ are then required so that the post-collision velocities can be calculated from eqn (2.5). To this end, a set of Cartesian coordinates x', y', and z' is introduced with x' in the direction of $\boldsymbol{c}_r$. The components of $\boldsymbol{c}_r^*$ along these axes are

$$c_r \cos\chi, \quad c_r \sin\chi \cos\varepsilon, \quad \text{and} \quad c_r \sin\chi \sin\varepsilon.$$

The direction cosines of x' are

$$u_r/c_r, \quad v_r/c_r, \quad \text{and} \quad w_r/c_r.$$

Since the orientation of the reference plane is arbitrary, the y'-axis may be chosen such that it is normal to the x-axis. The direction cosines of y' are then

$$0, \quad w_r(v_r^2 + w_r^2)^{-\frac{1}{2}}, \quad \text{and} \quad -v_r(v_r^2 + w_r^2)^{-\frac{1}{2}}$$

and those of z' follow as

$$(v_r^2 + w_r^2)^{\frac{1}{2}}/c_r, \quad -u_r v_r (v_r^2 + w_r^2)^{-\frac{1}{2}}/c_r, \quad \text{and} \quad -u_r v_r (v_r^2 + w_r^2)^{-\frac{1}{2}}/c_r.$$

The required expressions for the components of c_r^* in the original x-, y-, and z-coordinates are, therefore,

$$u_r^* = \cos\chi\, u_r + \sin\chi \sin\varepsilon\, (v_r^2 + w_r^2)^{\frac{1}{2}},$$

$$v_r^* = \cos\chi\, v_r + \sin\chi\, (c_r w_r \cos\varepsilon - u_r v_r \sin\varepsilon)/(v_r^2 + w_r^2)^{\frac{1}{2}}, \tag{2.22}$$

and

$$w_r^* = \cos\chi\, w_r - \sin\chi\, (c_r v_r \cos\varepsilon + u_r w_r \sin\varepsilon)/(v_r^2 + w_r^2)^{\frac{1}{2}}.$$

A *molecular model* is established through the definition of the force F or potential ϕ. As noted in Chapter 1, the force between real molecules is strongly repulsive at short distances and weakly attractive at larger distances. Models for analytical and numerical studies involve some degree of approximation. The simplest acceptable model is generally chosen, and most of these neglect the attractive component. A model may be said to be acceptable for a particular application if it leads to sufficiently accurate correlations between theory and experiment.

2.4 The inverse power law model

This model is sometimes called the point centre of repulsion model and is defined by

$$F = \kappa/r^{\eta}$$

or

$$\phi = \kappa/[(\eta-1)r^{\eta-1}]. \tag{2.23}$$

The ratio of the potential energy to the asymptotic kinetic energy may be written

$$\frac{\phi}{\frac{1}{2}m_r c_r^2} = \frac{2\kappa}{(\eta-1)r^{\eta-1}}m_r c_r^2 = \frac{2}{\eta-1}\left(\frac{W}{W_0}\right)^{\eta-1},$$

where W_0 is a second dimensionless impact parameter defined by

$$W_0 = b(m_r c_r^2/\kappa)^{1/(\eta-1)}. \tag{2.24}$$

Eqns (2.19) to (2.21) then show that the deflection angle is given by

$$\chi = \pi - 2\int_0^{W_1}[1-W^2-\{2/(\eta-1)\}(W/W_0)^{\eta-1}]^{-\frac{1}{2}}\,dW \tag{2.25}$$

where W_1 is the positive root of the equation

$$1-W^2-\{2/(\eta-1)\}(W/W_0)^{\eta-1} = 0.$$

Note that, for a given η, χ is a function of the dimensionless impact parameter W_0 only. This single parameter dependence of the deflection angle is the basic reason for the usefulness of the inverse power law model. The differential cross-section is a function of c_r and, for a fixed value of c_r, eqn (2.24) and its derivative may be substituted into eqn (2.11), to give

$$\sigma\,d\Omega = W_0\{\kappa/(m c_r^2)\}^{2/(\eta-1)}dW_0\,d\varepsilon. \tag{2.26}$$

For any finite value of η, the force field extends to infinity and the integral in eqn (2.14) for the total cross-section diverges.

The problem of an unbounded total collision cross-section σ_T is common to many classical models. Although σ_T is infinite, the vast majority of collisions are glancing collisions involving very slight deflections. It can be shown (Vincenti and Kruger 1965, Chap. IX, §8) that, when quantum effects are taken into account, the uncertainty principle does not allow these collisions to be properly defined. Therefore, when applying these models, a finite *cut-off* is a theoretical requirement as well as a practical necessity. Such a cut-off may be based either on the miss-distance b or the deflection angle χ. In most cases, the latter is to be preferred and, since χ is a function of W_0 for the inverse power law model, a deflection angle cut-off for this model may be applied through the specification of a maximum value $W_{0,m}$ of W_0. Then, for a fixed value of c_r, the total cross-section σ_T

may be obtained by integrating eqn (2.26) over all possible values of the impact parameters A_0 and ε, i.e.

$$\sigma_T = \int_0^{2\pi} \int_0^{W_{0,m}} W_0 \{\kappa/(m_r c_r^2)\}^{2/(\eta-1)} \, dW_0 \, d\varepsilon .$$

or

$$\sigma_T = \pi W_{0,m}^2 \{\kappa/(m_r c_r^2)\}^{2/(\eta-1)} = \pi W_{0,m}^2 \{\tfrac{1}{2}\kappa/E_t\}^{2/(\eta-1)}. \tag{2.27}$$

However, since the value of $W_{0,m}$ is arbitrary, this expression is not suitable for setting the effective collision frequency or mean free path.

The calculation of the transport properties, such as the coefficient of viscosity, will be found to involve the differential rather than the total cross-section. A nominal total cross-section for a particular model may therefore be defined as the cross-section of the hard sphere model that matches the coefficient of viscosity for that model. This procedure effectively fixes the value of the constant κ. However, eqn (2.27) shows that the total cross-section of an inverse power law molecule is inversely proportional to $c_r^{4/(\eta-1)}$, and the hard sphere model is preferably replaced by either the VHS model (see §2.6) or the VSS model (see §2.7) with the same dependence on the relative speed.

The nominal cross-section established through the above procedure is not to be confused with the viscosity cross-section σ_μ. This is defined by

$$\sigma_\mu = \int_0^{4\pi} \sin^2\chi \, \sigma \, d\Omega$$

or, from eqn (2.12),

$$\sigma_\mu = 2\pi \int_0^{\pi} \sigma \sin^3\chi \, d\chi \tag{2.28}$$

and its name is derived from the fact that it is a convergent integral that is encountered in the Chapman–Enskog theory for the coefficient of viscosity. Also, it may be seen from Fig. 2.1(c) that the component of post-collision velocity in the direction of the pre-collision velocity is $(1-\cos\chi)c_r$. This leads to the momentum transfer cross-section σ_M defined by

$$\sigma_M = \int_0^{4\pi} (1-\cos\chi) \, \sigma \, d\Omega$$

or

$$\sigma_M = 2\pi \int_0^{\pi} \sigma(1-\cos\chi) \sin\chi \, d\chi . \tag{2.29}$$

The momentum cross-section appears in the Chapman–Enskog theory for the diffusion coefficients, and may be referred to by the alternative name of *diffusion cross-section*.

2.5 The hard sphere model

The hard sphere model that was employed in much of the discussion in Chapter 1 may be regarded as the special case of the inverse power law model with $\eta = \infty$. As shown in Fig. 2.4, this force becomes effective at

$$r = \tfrac{1}{2}(d_1 + d_2) = d_{12}$$

and the apse line is the line through the centres of the spheres. Therefore,

$$b = d_{12} \sin\theta_A = d_{12} \cos(\tfrac{1}{2}\chi)$$

and

$$|\,db/d\chi\,| = \tfrac{1}{2} d_{12} \sin(\tfrac{1}{2}\chi)$$

so that eqn (2.13) gives

$$\sigma = d_{12}^2/4. \tag{2.30}$$

This equation shows that σ is independent of χ and that the scattering from hard sphere molecules is isotropic in the centre of mass frame of reference. That is, all directions are equally likely for $\boldsymbol{c}_r^*$, as can be seen from purely geometric considerations. The total collision cross-section is, as expected,

$$\sigma_T = \int_0^{4\pi} \sigma \, d\Omega = \pi d_{12}^2. \tag{2.31}$$

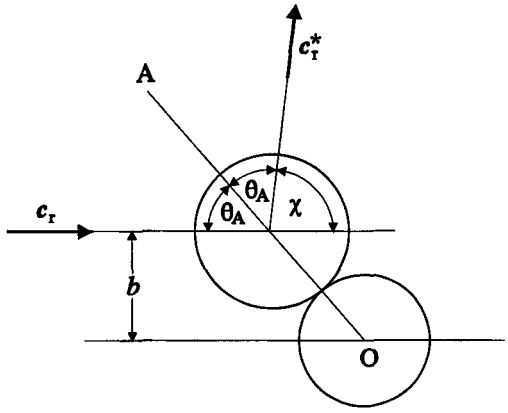


Fig. 2.4 Collision geometry of hard sphere molecules.

The viscosity cross-section is defined in eqn (2.28) and, for the hard sphere model, the constant cross-section may be taken outside the integral, to give

$$\sigma_\mu = \frac{2}{3}\sigma_T. \tag{2.32}$$

The momentum or diffusion coefficient that is defined in eqn (2.29) is given similarly as

$$\sigma_M = \sigma_T. \tag{2.33}$$

2.6 The variable hard sphere (VHS) model

In addition to the finite total cross-section, the hard sphere model has the advantage of easily calculated collision mechanics because of the isotropic scattering in the centre of mass frame of reference. Its disadvantages are that this scattering law is not realistic and that the cross-section is independent of the relative translational energy $E_t = \frac{1}{2}m_r c_r^2$ in the collision. Other than at extremely low temperatures, the effective cross-section of real molecules decreases as c_r and E_t increase. The rate of decrease is directly related to the change of the coefficient of viscosity with temperature. The viscosity coefficient of the hard sphere model is proportional to the temperature to the power of 0.5 and a variable cross-section is required to match the powers of the order of 0.75 that are characteristic of real gases. It has been found that the consequences of changes in the molecular model in analytical and numerical studies of gas flows correlate strongly with this change in the effective cross-section. By contrast, the observable consequences of changes in the scattering law are generally very small.

These observations led to the introduction (Bird 1981) of the variable hard sphere, or VHS, molecular model. This is a hard sphere molecule with a diameter d that is a function of c_r. The function is generally, but not necessarily, a simple inverse power law υ, so that

$$d = d_{ref}(c_{r,ref}/c_r)^{\upsilon}, \tag{2.34}$$

where the subscript $_{ref}$ denotes reference values. It will be shown in §3.5 that, if this power law follows eqn (2.27), the variation of the coefficient of viscosity with temperature is the same as that in the corresponding inverse power law model. The VHS model for a particular gas may be defined (§4.3) by the effective diameter at a particular reference temperature.

Because the VHS model combines a finite cross-section with a realistic temperature exponent of the coefficient of viscosity, it has permitted the definition (Bird 1983) of a mean free path and a Knudsen number that account for the real gas temperature exponent of the coefficient of viscosity. The deflection angle is given by the hard sphere result

$$\chi = 2\cos^{-1}(b/d) \tag{2.35}$$

and, while d is a function of c_r, there is no coupling between c_r and χ. This means that σ may again be taken outside the integral in eqns (2.28) and (2.29), and the viscosity and momentum cross-sections are again given by eqns (2.32) and (2.33).

2.7 The variable soft sphere (VSS) model

As noted above, the cross-section in VHS model is determined from the viscosity coefficient, but the ratio of the momentum to the viscosity cross-section follows the hard sphere value. This is a deficiency in the model because the ratio varies with the power law in the case of the inverse power law model, and the effective real gas values are different from the hard sphere value. In response to this problem, Koura and Matsumoto (1991 and 1992) introduced the variable soft sphere (or VSS) model in which the diameter varies in the same way as in the VHS model, but the deflection angle is given by

$$\chi = 2\cos^{-1}\{(b/d)^{1/\alpha}\}. \tag{2.36}$$

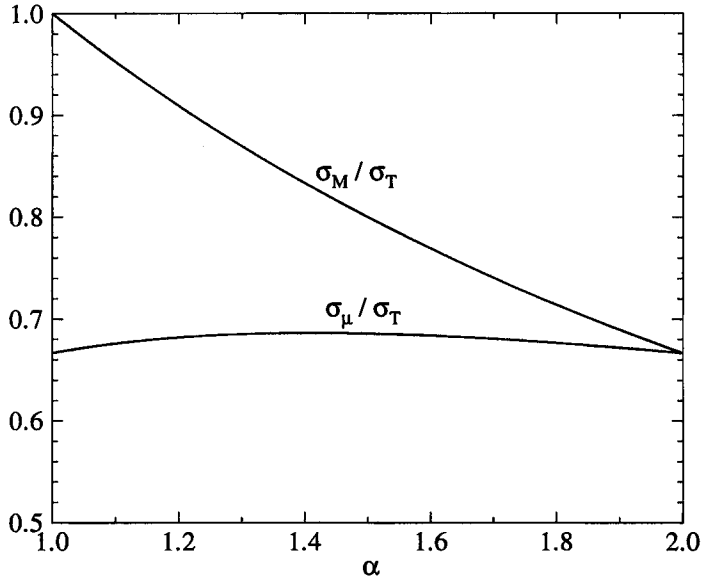


Fig. 2.5 The viscosity and momentum cross-sections as functions of the exponent in the VSS model.

The cross-section may again be taken outside the integral and it may be shown that the total cross-section σ_T remains equal to πd^2. The viscosity and momentum cross-sections become

$$\sigma_\mu = \frac{4\alpha}{(\alpha+1)(\alpha+2)}\sigma_T. \tag{2.37}$$

and

$$\sigma_M = \frac{2}{(\alpha+1)}\sigma_T. \tag{2.38}$$

Eqn (A2) shows that the Schmidt number, which is essentially the ratio of the viscosity to the diffusion, can be expressed as a function of the temperature exponent of the viscosity cofficient ω and α. The values of α generally lie between 1 and 2. Figure 2.5 shows that the viscosity cross-section is barely affected, while the momentum or diffusion coefficient depends strongly on α. A typical variation of the deflection angle with the impact parameter b is shown in Fig. 2.6 for the VHS, VSS and inverse ninth power law models. The normalization is such that all three models lead to the same coefficient of viscosity, but only the VSS and inverse power law models have the same diffusion coefficients. The VSS model leads to near 90° deflection angles over a wider range of impact parameters.

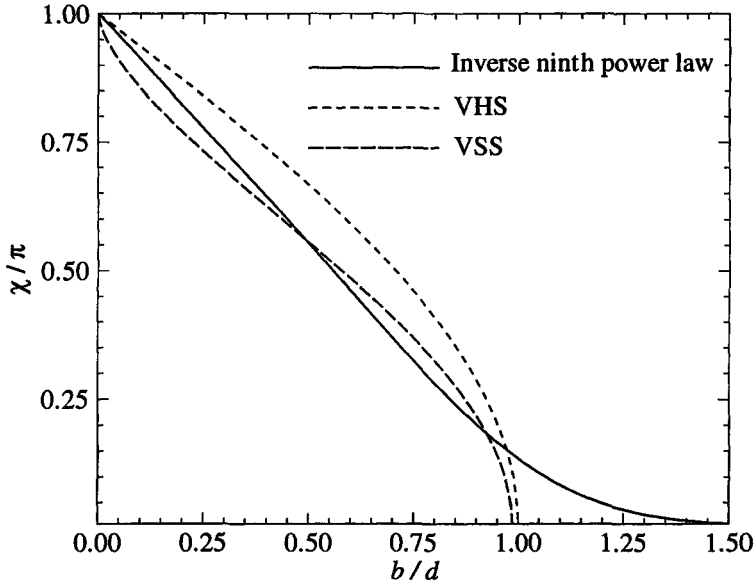


Fig. 2.6 The deflection angle in a collision as a function of the miss-distance impact parameter.

2.8 The Maxwell model

This is the special case of the inverse power law model with $\eta=5$. We then have

$$W_1 = W_0^2[\{1+(2/W_0^4)\}^{1/2}-1]^{1/2}$$

for the dimensionless distance of closest approach and eqn (2.25) may be integrated. The result is

$$\chi = \pi - [2/\{1+(2/W_0^4)\}^{1/4}]\, K\,[1/2 - 1/2\{1+(2/W_0^4)\}^{-1/2}], \tag{2.39}$$

where

$$K(\alpha) = \int_0^{\pi/2} (1-\alpha\sin^2 y)^{-1/2}\,dy$$

is the complete elliptic integral of the first kind. Eqn (2.26) for the differential cross-section becomes

$$\sigma\,d\Omega = \frac{W_0}{c_r}\left(\frac{\kappa}{m_r}\right)^{1/2} dW_0\,d\varepsilon. \tag{2.40}$$

We saw in §1.3 that the collision probability for a pair of molecules is proportional to the product of the cross-section and the relative speed. Eqn (2.40) therefore shows that the collision probability for a particular molecule in a Maxwellian gas is independent of velocity.

A gas of Maxwell molecules has a coefficient of viscosity that is linearly proportional to the temperature. This is unrealistic for real gases. The model is widely used for analytical studies but, in almost all cases, this is because all molecules have the same collision probability, and the analysis would not be possible if the collision probability was dependent on the relative speed. The model can be regarded as the limiting case of a 'soft' molecular model, while the hard sphere model may be regarded as the limiting case of a 'hard' molecular model.

2.9 Models that include the attractive potential

All the preceding models lead to a coefficient of viscosity that is either constant or is proportional to a *constant* power of the temperature. The addition of a long-range attractive potential to the inverse (repulsive) power law model leads to a better representation of the potential curve of real molecules. It also leads to a temperature-dependent power law for the variation of the coefficient of viscosity with temperature. The effect of the attractive potential is significant at temperatures well below the standard temperature, but is generally not significant at higher temperatures.

The *square-well model* adds a uniform attractive potential to the hard sphere model, while the *Sutherland model* adds an inverse power law attractive component to the hard sphere model. The more realistic models combine more general attractive and repulsive potentials. The best known of the attractive–repulsive models is the *Lennard–Jones potential* which

adds an inverse power law attractive component to the inverse power law model. The equation for the potential is

$$\phi = \frac{\kappa}{(\eta-1)r^{\eta-1}} - \frac{\kappa'}{(\eta'-1)r^{\eta'-1}}. \tag{2.41}$$

This model has four adjustable parameters and these allow the transport coefficients to be matched with the experimental values over a wide temperature range. The Lennard–Jones 12-6 model with η=13 and η'=7 has been widely used. The *Exp-6* model replaces the repulsive power law with a more complex exponential term. A detailed discussion and evaluation of these and other models can be found in Hirschfelder, Curtiss, and Bird (1954).

2.10 The generalized hard sphere (GHS) model

The GHS model (Hasssan and Hash 1993) is an extension of the VHS and VSS models. It bears the same relationship to the Lennard-Jones class of models as the conventional VHS or VSS models bear to the inverse power law model. The scattering distribution is that of the hard or soft sphere, but the variation of the total cross-section as a function of the relative translational energy mimics that of the corresponding attractive-repulsive potential. It is implemented through the parameters that describe the intermolecular potentials of the form of eqn (2.34), and can therefore make use of the existing database that has been built up from the measured transport properties of real gases.

2.11 General comments

It should be kept in mind that the justification for using any molecular model is that complete information on the differential cross-sections of the real molecules does not exist, would not affect the results, or renders the calculation intractable. The latter justification becomes more irrelevant as computer capability increases and more powerful numerical methods are introduced. Comprehensive theoretical and experimental information on cross-sections is available for some monatomic gases at low temperatures. Chatwani (1977) has used this data instead of a classical molecular model in studies of expansion flows of helium and neon. The cross-sections at very low temperatures are subject to significant quantum effects and can be complex functions of the relative velocity. The direct use of quantum information on the differential cross-sections is physically realistic, but the same results could well be obtained through the incorporation of the resulting total cross-section into the VHS model.

A model that reproduces the overall effects in a gas flow of some physical feature of real molecules, but does not explicitly incorporate that feature, is said to be a *phenomenological model*. The use of a VHS model, rather than one of the models with a more realistic scattering law, to produce the required temperature dependence of the transport properties is an example

of phenomenological modelling. The VHS model was empirically based on computational studies which showed that the variations in cross-section produce effects that are incomparably more important than those due the scattering laws. Most phenomenological methods are, however, based on analytical studies, rather than observations of numerical studies.

Later chapters deal with the *inelastic collisions* of molecules with internal degrees of freedom. The models for these that were developed in the context of classical kinetic theory will be found to have restrictions that render them incapable of providing an adequate representation of real gases. Also, the number of impact parameters increases with the number of degrees of freedom, and the amount of information that would be required for a complete description of the differential cross-sections becomes prohibitively large. As the models are extended to deal with the inelastic aspects of real molecules, the practical models will be found to be increasingly phenomenological in nature.

References

BIRD, G.A. (1981). Monte Carlo simulation in an engineering context. *Progr. Astro. Aero.* **74,** 239–255.

BIRD, G.A. (1983). Definition of mean free path for real gases. *Phys. Fluids* **26,** 3222-3223.

CHATWANI, A.U. (1977). Monte Carlo simulation of nozzle beam expansion. *Progr. Astro. Aero.* **51,** 461-475.

HASSAN, H.A. and HASH, D.B. (1993). A generalized hard-sphere model for Monte Carlo simulations. *Phys. Fluids A* **5**, 738–744.

HIRSCHFELDER, J.O., CURTISS, C.F., and BIRD, R.B. (1954). *The molecular theory of gases and liquids*. Wiley, New York.

KOURA, K. and MATSUMOTO, H. (1991). Variable soft sphere molecular model for inverse-power-law or Lennard-Jones potential. *Phys. Fluids A* **3**, 2459-2465.

KOURA, K. and MATSUMOTO, H. (1992). Variable soft sphere molecular model for air species, *Phys. Fluids A* **4**, 1083-1085.

VINCENTI, W.G. and KRUGER, C.H. (1965). *Introduction to physical gas dynamics.* Wiley, New York.